

VELMÈRE PRO AUDIT REPORT

Tornado Cash Router (0xd90e2f925da726b50c4ed8d0fb90ad053324f31b)

VELMERE SECURITY AUDIT REPORT: TORNADO CASH ROUTER

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: undefined | Tier: PRO | Application Surface: CANONICAL

Created At: 2026-09-09T19:29:42.384Z

VERDICT SUMMARY:

HIGH REGULATORY RISK (84/100)

Release Decision: BLOCKED | Verification Status: PARTIALLY_VERIFIED

STOP-SELL DIRECTIVE: CRITICAL_RISK_THRESHOLD: Score exceeds maximum institutional tolerance

Confidence Score: 99/100 Evidence Coverage: 98%

Merkle Root: sha256:4a09402e5af0268ba11a0e98c8546ef3f32b0446f1de7aa971c4ba8ad18b39ff

CRITICAL NOTICE: Contract is designated on international OFAC sanctions lists. Direct interaction triggers immediate automated compliance freezing across regulated institutions.

AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Tornado Cash Router

Contract Address: 0xd90e2f925da726b50c4ed8d0fb90ad053324f31b

Network / Chain ID: Ethereum Mainnet (ID: 1)

Token / Symbol: TORN-ROUTER

Proxy Pattern: Immutable (No Proxy)

Compiler Version: solc 0.7.6

SOURCE CODE VERIFICATION & BYTECODE DECOMPILEATION

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

Findings & Vulnerability Analysis:

CRITICAL VLM-TORN-01: OFAC Sanctions List Collision

Category: Regulatory Compliance | State: recommended

Recommendation: Institutional entities must block routing through this contract to maintain regulatory compliance.

PERMISSION & ROLE GOVERNANCE MAP (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Deep analytical map of privileged roles, owner powers, and timelock bounds

* Relayer Authorization	Decentralized cryptographic relayer registry	VERIFIED
* Zero-Knowledge Verification	Groth16 zk-SNARK proof verification	VERIFIED

Detailed role map traces all execution paths back to verifiable on-chain controllers.

Findings & Vulnerability Analysis:

HIGH VLM-TORN-P01: Automated AML Taint Propagation

Category: Taint Analysis | State: open

Evidence: Deterministic AML cluster taint detection

Recommendation: Do not route treasury or client funds through this router.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Token distribution curve, whale clusters, and verifiable DEX lock proofs

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* Deposit Anonymity Pool	Historically shielded pool balances	NEUTRAL
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Liquidity analysis examines AMM pairing architecture, lock mechanisms, and transaction fee behavior.

ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

* Oracle Dependency	Pricing feeds inspected in bytecode	VERIFIED
* Flash Loan Attack Vector	State manipulation resilience assessed	VERIFIED
* Reentrancy Protection	CEI pattern / Reentrancy guard verified	VERIFIED

STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

[LOCKED] [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

[LOCKED] [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): PROPERTY P VERIFIED UNDER ASSUMPTIONS A FOR MODEL M
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

[LOCKED] [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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